

# **The Big Beautiful Quantum Box (BBQB): CSM, Quantum Logic, and Some Related Snippets**

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## **Part I**

# **CSM Framework vs. Quantum Logic**

# CSM vs. Quantum Logic

|                            | Quantum Logic<br>Birkhoff–von Neumann                           | CSM<br>Contexts–Systems–<br>Modalities                               |
|----------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Primitive notion</b>    | Propositions: yes/no tests, or<br>thogonal projection operators | Modalities: certain, repeat-<br>able phenomena                       |
| <b>Structure</b>           | Orthomodular lattice of pro-<br>jectors                         | Contexts: complete sets of $N$<br>mutually exclusive modalities      |
| <b>Role of context</b>     | Boolean subalgebras or blocks                                   | Explicit and primary; a modal-<br>ity exists <i>within</i> a context |
| <b>Ontology</b>            | Properties of the system                                        | Properties of system and con-<br>text jointly                        |
| <b>Probabilities</b>       | Via Gleason                                                     | Via Gleason/Uhlhorn                                                  |
| <b>Goal</b>                | Algebraic logic of qm                                           | Derive Hilbert space from de-<br>siderata                            |
| <b>Common<br/>elements</b> | Gleason’s “intertwining” of<br>observables across bases         | Extravalence: one modality<br>appearing in many contexts             |

# Desiderata

- In any context there are exactly  $N$  mutually exclusive modalities ( $N$  fixed by the system).
- The certainty of a modality can carry over between contexts (extravalence)—but in general only *probabilistically*, since  $N$  cannot increase.
- Contexts and modalities change continuously (connected groups of context transformations).

*Hilbert space is a formalism compatible with desiderata.*

## **Part II**

# **Matrix Pencils, Peres–Mermin Square, and GHZ Arguments**

## Matrix Pencils: Motivation

- Operator-valued arguments can hide the underlying measurement contexts: orthonormal bases or projection operators.
- Non-degenerate operators reveal their eigensystems directly.
- Degenerate commuting operators require a systematic way to extract the common maximal-resolution context.

*Aim: expose the context behind a commuting family of observables.*

## Matrix Pencils: Construction

Let  $A_1, \dots, A_I$  be mutually commuting degenerate operators. Form

$$P = \sum_{i=1}^I a_i A_i,$$

where the  $a_i$  are generic real scalars.

- $P$  commutes with every  $A_i$ .
- For generic coefficients,  $P$  removes accidental degeneracies.
- Diagonalizing  $P$  gives the simultaneous eigensystem shared by the commuting observables.

# Peres–Mermin Square

- The Peres–Mermin square contains nine dichotomic Pauli observables in a  $3 \times 3$  grid.
- Each row and each column is a commuting set, giving six measurement contexts.
- Applying matrix pencils to these commuting sets reveals their common quantum-logical structure.

*The square is a compact operator form of contextuality; the pencil extracts the underlying contexts.* doi:

10.1103/PhysRevA.110.012215



## From PM Square to KS Configurations

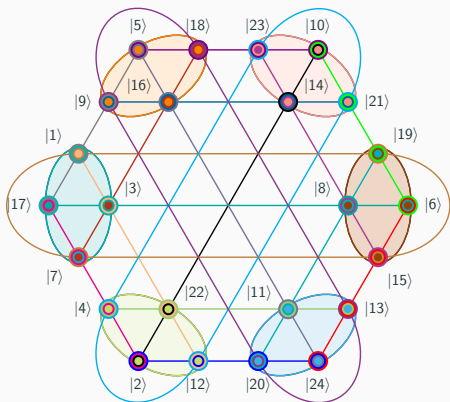
- The extracted eigensystems yield a **24–24 configuration**: 24 propositions arranged into 24 contexts.
- This completes the well-known **18–9 configuration** of Cabello, Estebaranz, and García-Alcaine.
- No classical two-valued truth assignment exists.

**Conclusion: strong quantum contextuality.**

# Peres-Mermin Square

$$\begin{pmatrix} \sigma_z \mathbb{1}_2 & \mathbb{1}_2 \sigma_z & \sigma_z \sigma_z \\ \mathbb{1}_2 \sigma_x & \sigma_x \mathbb{1}_2 & \sigma_x \sigma_x \\ \sigma_z \sigma_x & \sigma_x \sigma_z & \sigma_y \sigma_y \end{pmatrix}$$

$\equiv$



# Tripartite GHZM Argument

- Mermin's GHZM version uses four commuting three-partite operators.
- A matrix pencil reveals one context of eight (entangled) states.  
doi: 10.1007/s10701-021-00515-z
- A parity argument gives a state-dependent contradiction.

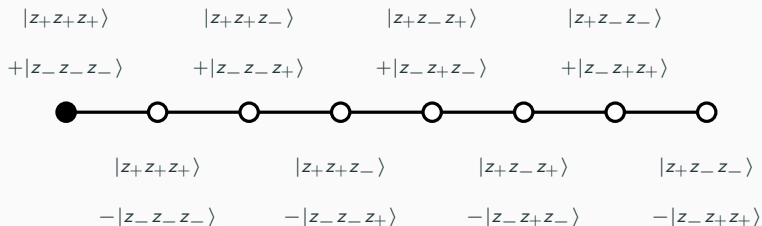
**Quantum prediction:  $-1$**

**Classical prediction:  $+1$**

# GHZ Observables and States

$\sigma_x \sigma_x \sigma_x$ ,  $\sigma_x \sigma_y \sigma_y$ ,  $\sigma_y \sigma_x \sigma_y$ , and  $\sigma_y \sigma_y \sigma_x$ .

$\equiv$



## Bipartite GHZM from the PM Square

- A complete GHZ-type contradiction can be obtained with two particles.
- Use non-local observables from the last row and column of the Peres–Mermin square.
- Representative products include

$$(\sigma_z \otimes \sigma_x)(\sigma_x \otimes \sigma_z), \quad (\sigma_x \otimes \sigma_x)(\sigma_z \otimes \sigma_z).$$

- The associated matrix pencil yields the Bell basis as a single four-dimensional context.

doi: 10.1103/PhysRevA.110.012215

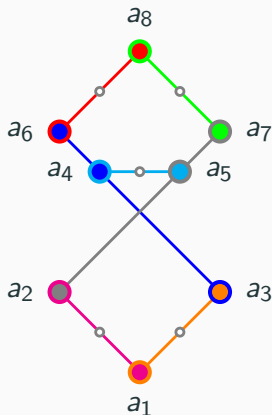
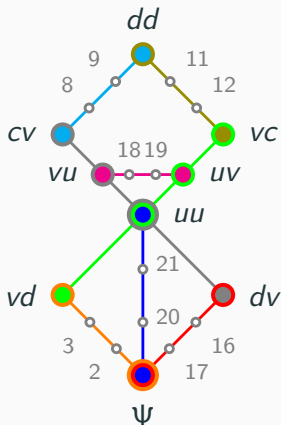
## Bipartite GHZM: Parity Contradiction

- Classical “elements of reality” occur an even number of times.
- Therefore any classical product assignment predicts  $+1$ .
- Quantum mechanically, the product equals  $\langle \mathbb{1}_2 \otimes (-\mathbb{1}_2) \rangle = -\langle \mathbb{1}_4 \rangle = -1$ .

**Classical:**  $+1$       **Quantum:**  $-1$

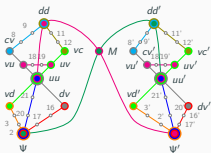
# The Specker bug: “Hardy’s wonderful trick” [Mermin], also called “Hardy’s beautiful example” [Penrose]

TIFS:  
 $1 \longrightarrow 0$

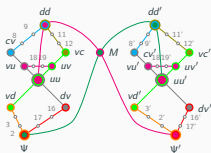


**Reference:** A. Cabello, J. M. Estebaranz, and G. García-Alcaine. *Bell-Kochen-Specker theorem: A proof with 18 vectors*. Physics Letters A, 212(4): 183-187, 1996. DOI: 10.1016/0375-9601(96)00134-X. arXiv: quant-ph/9706009.

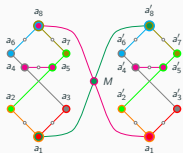
## Specker Bug Combo: nonseparable set of two-valued states – strong contextuality



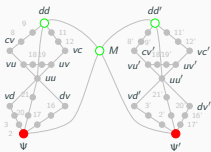
(a)



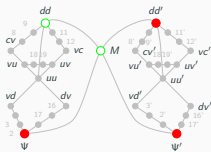
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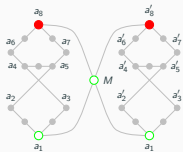
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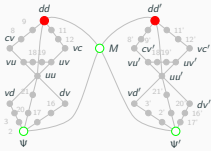
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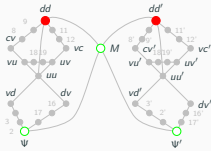
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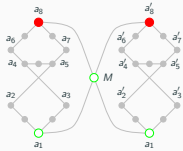
(f)



(g)



(h)



(i)



## **Part III**

# **The Geometric Mechanism of Decoherence**

# Geometry Behind Decoherence

- Decoherence requires dynamics to correlate system and environment.
- Its effectiveness is also kinematically supported by high-dimensional geometry.
- In a high-dimensional Hilbert space, almost all random pure states are nearly orthogonal.

doi:10.48550/arXiv.2605.03807

## Typical Overlap in Dimension $d$

Let  $|\psi\rangle = (\psi_1, \dots, \psi_d)$  be a random unit vector, and let  $\mathbb{E}$  denote the mathematical expectation and  $\mathbb{P}$  denote probability:

$$\sum_{i=1}^d |\psi_i|^2 = 1.$$

- By symmetry,  $\mathbb{E}|\psi_1|^2 = \dots = \mathbb{E}|\psi_d|^2$ .
- Hence

$$d \mathbb{E}|\psi_1|^2 = 1, \quad \mathbb{E}|\psi_1|^2 = \frac{1}{d}.$$

- Aligning a fixed state  $|\phi\rangle$  with the first vector of the Cartesian standard basis  $|e_1\rangle$  gives

$$\mathbb{E} |\langle \phi | \psi \rangle|^2 = \frac{1}{d}.$$

# Exponential Tail

Lévy's lemma, Johnson–Lindenstrauss lemma: For a fixed  $|\phi\rangle$  and random  $|\psi\rangle \in \mathbb{C}^d$ ,

$$\mathbb{P}\left\{|\langle\phi|\psi\rangle|^2 \geq \varepsilon\right\} = (1 - \varepsilon)^{d-1} \leq e^{-(d-1)\varepsilon}.$$

- Large dimension makes sizeable overlaps exponentially rare.
- This is the geometric reservoir behind quasi-orthogonality.

- Geometry supplies a vast quasi-orthogonal reservoir.
- The Hamiltonian selects and populates the pointer basis.
- Overlap amplitudes scale exponentially, so macroscopic alternatives remain distinct for all practical purposes.

*No Schrödinger “quantum jelly”: high-dimensional Hilbert space separates orthogonal states FAPP.*

*Thank you for your attention!*